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Overview

Topic	Description
Module Purpose	Handles AI image classification and OCR-based ID card processing
Main Features	Image classification, OCR extraction, validation, comparison
Technologies	ML.NET, Tesseract OCR, ImageSharp

Training Notes & Video Reference

Item	Details
YouTube Video	ML.NET Image Classification Training Guide
Common Training Issue	Model freezes during training without visible error
Manual Model File	https://aka.ms/mlnet-resources/meta/resnet_v2_50_299.meta
Required Path	C:\Users\hp\AppData\Local\Temp\MLNET\resnet_v2_50_299.meta
Best Practice	Keep project outside C:\ drive

OCR Project Overview

Aspect	Description
Supported Files	Images and PDF documents
Languages	Arabic and English
Main Libraries	SixLabors.ImageSharp, Tesseract OCR

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Image Processing Libraries

Library	Purpose
SixLabors.ImageSharp	Image sharpening, grayscale conversion, cropping
Tesseract OCR	Text extraction from images (Arabic & English)

Tesseract Language Data

Language File	Description	Source
ara.traineddata	Arabic OCR language file	tessdata / tessdata_best
eng.traineddata	English OCR language file	tessdata / tessdata_best

Place the downloaded files inside a folder named **tessdata** and include it in the project using Tesseract.

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IDCardExtractedDataVM Properties

Property	Type	Description
lable	string?	AI-predicted document type (e.g., ID)
Probability_double	double?	Classification confidence (0–100)
ProbabilityString	string?	Formatted probability percentage
doneAI_bool	bool?	AI classification success flag
doneOCR_bool	bool?	OCR extraction success flag
doneValidation_bool	bool?	Data validation success flag
textExtracted	string?	Raw OCR extracted text
matchIDNumber	string?	Extracted ID number
matchFullEnName	string?	Extracted English name
matchFullArName	string?	Extracted Arabic name
matchBirth	string?	Extracted date of birth
matchExpiryDate	string?	Extracted ID expiry date
matchGender	string?	Extracted gender
matchNationality	string?	Extracted nationality

IDClassificationController — Processing Flow

Step	Description
1	Validate uploaded file
2	Run AI image classification
3	Check confidence threshold (≥ 90%)
4	Convert image to grayscale
5	Extract all OCR text fields
6	Populate IDCardExtractedDataVM
7	Optional data comparison & validation

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CompareService Capabilities

Method	Purpose
LevenshteinDistance	Calculates character edit distance
SimilarityPercentage	Returns similarity percentage (0–100)

ID Classification Page — UI Flow

Section	Description
Upload Form	User uploads ID image
Validation	Client-side and server-side file checks
Result Display	Shows classification label, confidence, and extracted fields
Error Handling	Friendly messages shown for failures

- ✓ AI + OCR integrated workflow
- ✓ Structured validation & comparison
- ✓ Multilingual ready (Arabic / English)
- ✓ Extensible for future AI enhancements